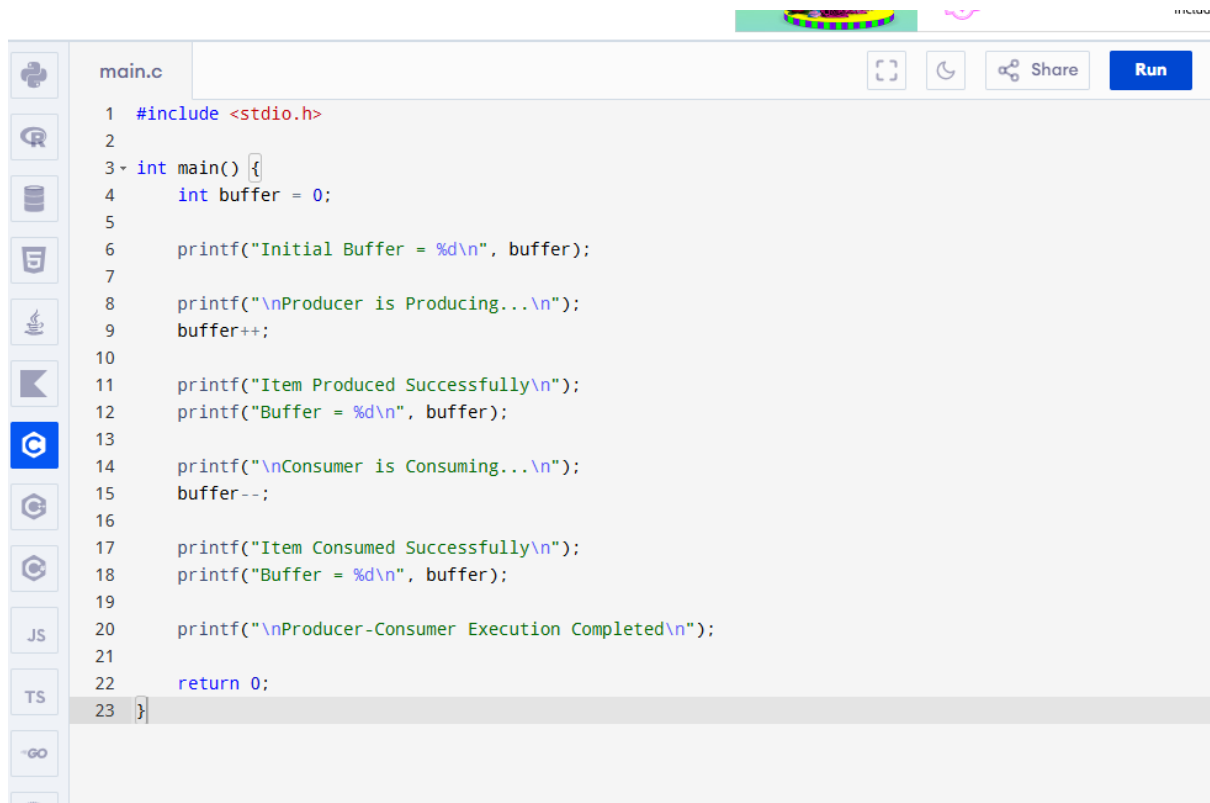


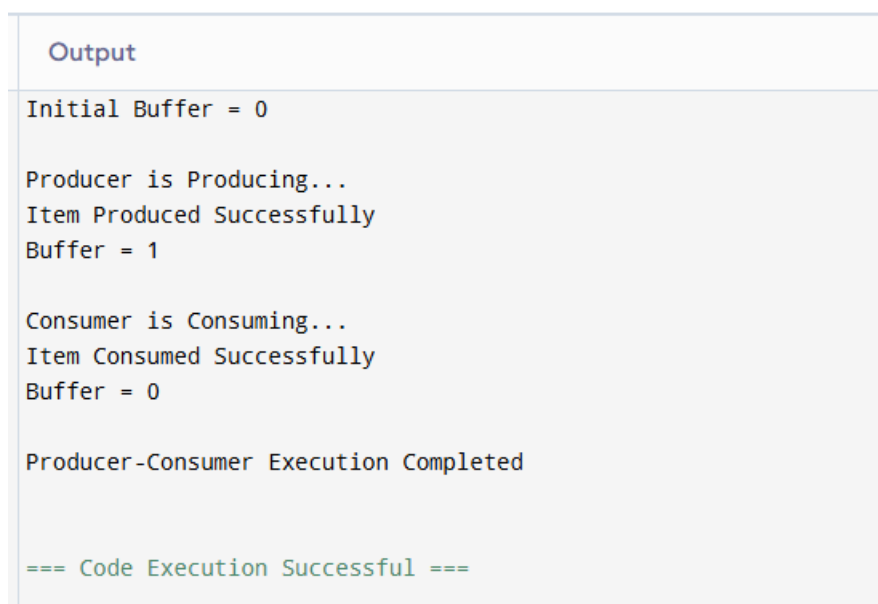
18 Construct a C program to simulate producer-consumer problem using semaphores.

Code:

A screenshot of a code editor window showing a C program. The editor has a sidebar on the left with icons for various languages and a top bar with a 'Run' button and a 'Share' button. The code is as follows:

```
1 #include <stdio.h>
2
3 int main() {
4     int buffer = 0;
5
6     printf("Initial Buffer = %d\n", buffer);
7
8     printf("\nProducer is Producing...\n");
9     buffer++;
10
11    printf("Item Produced Successfully\n");
12    printf("Buffer = %d\n", buffer);
13
14    printf("\nConsumer is Consuming...\n");
15    buffer--;
16
17    printf("Item Consumed Successfully\n");
18    printf("Buffer = %d\n", buffer);
19
20    printf("\nProducer-Consumer Execution Completed\n");
21
22    return 0;
23 }
```

Output:

A screenshot of the output window of the code editor. It shows the following text:

```
Output

Initial Buffer = 0

Producer is Producing...
Item Produced Successfully
Buffer = 1

Consumer is Consuming...
Item Consumed Successfully
Buffer = 0

Producer-Consumer Execution Completed

=== Code Execution Successful ===
```